

Exp 2: Monoalphabetic Substitution Cipher

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
int main() {
```

```
    char text[100];
```

```
    char key[27] = "QWERTYUIOPASDFGHJKLZXCVBNM";
```

```
    int i, pos;
```

```
    printf("Enter text: ");
```

```
    scanf("%s", text);
```

```
    for(i = 0; text[i] != '\0'; i++) {
```

```
        if(isalpha(text[i])) {
```

```
            pos = toupper(text[i]) - 'A';
```

```
            text[i] = key[pos];
```

```
        }
```

```
    }
```

```
    printf("Encrypted text: %s\n", text);
```

```
    for(i = 0; text[i] != '\0'; i++) {
```

```
        for(pos = 0; pos < 26; pos++) {
```

```
            if(text[i] == key[pos]) {
```

```
                text[i] = 'A' + pos;
```

```
                break;
```

```
            }
```

```
        }
```

```
    }
```

```
printf("Decrypted text: %s\n", text);
```

```
return 0;
```

```
}
```

```
1  #include <stdio.h>
2  #include <ctype.h>
```

```
Enter text: hello
Encrypted text: ITSSG
Decrypted text: HELLO
```

```
-----
Process exited after 3.188 seconds with return value 0
Press any key to continue . . . |
```